

Case Study: Mapping Water in the Sundarbans — with SatQuery AI

A real place, real satellite data, real answers — end to end.

The scene

We gave SatQuery AI a pair of real satellite images of the **Hooghly river estuary in the Sundarbans delta, West Bengal** (88.13°E, 22.31°N — one of the world's largest river deltas, where flood and water monitoring genuinely matter):

Optical (Sentinel-2)



Radar (Sentinel-1 SAR)



- **Left (optical):** what a normal camera-in-space sees — the wide river channel in blue-green, vegetated delta land around it.
- **Right (SAR):** a radar image — it sees through clouds and at night. Water shows up dark because calm water reflects radar away from the satellite.

What we asked — and what happened

1. "Is there water in this scene?" (optical + SAR together, 11.8 seconds)

SatQuery ran **two independent specialists** and compared them:

- The **SAR tool** found water covering **54.7%** of the scene
- The **optical tool** found water covering **53.3%** of the scene
- Their masks agree at **IoU 0.89** — two different sensors, looking at physics from completely different angles, landed on almost the same map



Agreement map: green = both sensors agree it's water — the river channel traced exactly.

The narrated answer — written by the AI voice, but only from verified evidence:

"Water is confirmed in this scene, with both SAR and optical tools detecting it... The two masks agree strongly, with an IoU of 0.892."



Water mask overlaid on the optical image — the overlay hugs the riverbank.

2. "How big is the water body?" (optical alone, 22 seconds)

- **3.49 km²** — computed from real pixel size (each pixel = 10 m on the ground), not estimated by the language model
- 53.25% of the scene, concentrated in the north-east

Every image product was also exported as a **georeferenced GeoTIFF** ([water_sar_mask.tif](#), [water_optical_mask.tif](#), [agreement_map_live.tif](#), [water_mask.tif](#)) — meaning the outputs drop straight into professional GIS software (e.g., QGIS) correctly aligned to the real world. This is what turns a demo into an analyst-grade artifact.

3. The honesty moment (why this demo is different)

Two things happened that most demos would hide:

- **Co-registration check:** the system verified the two images are aligned — their recorded ground positions match exactly (offset **0.0 m**). But the pixel-level texture check came back "*inconclusive (weak peak)*" — radar and camera textures don't always correlate. The system **reported both honestly** instead of silently claiming a perfect alignment.
- **Two model architecture, one disagreement:** we asked "how many boats are visible?" The **domain-adapted model** (trained by us on remote-sensing data) answered **1**. The **frozen narrator** (untouched base model) answered **0**. The system **disclosed both answers with their model identities** rather than silently picking one — every claim in the trace carries which model said it and the exact file hash of that model.

What this demonstrates

Capability	Shown here
Understands plain-English questions	All queries were natural language
Multi-sensor reasoning	Optical + radar cross-validated, not blended blindly
Real measurements	3.49 km ² from actual ground-sample distance
Auditable	Every number traces to a named tool/model + file hash
Honest under uncertainty	Withheld/caveated where evidence was weak
Analyst-grade output	Georeferenced GeoTIFFs, usable in QGIS directly
Runs offline	Both AI models ran locally on a laptop — zero internet

Files in this folder

- `optical.png` · `sar_vv.png` — the two input views
- `overlay_live.png` — detected water over the optical image
- `agreement_map_live.png` — where the two sensors agree/disagree
- `*.tif` — georeferenced export artifacts (EPSG:32645)
- `trace*.json` — the full execution records (every tool call, claim, hash)

Appendix — the terms, in plain language

- **Optical image:** a photo from a satellite — like a camera in space.
- **SAR (Synthetic Aperture Radar):** an image made by bouncing radar pulses off the ground. Works through clouds and at night; water looks dark.
- **Backscatter (dB):** how strongly the ground bounces the radar back. Smooth water ≈ very low (dark).
- **IoU (Intersection over Union):** how much two outlines overlap, 0–1. 0.89 here = the two sensors' water maps nearly coincide.
- **GSD (Ground Sample Distance):** real-world size of one pixel — 10 m here, which is what lets us say "3.49 km²" honestly.
- **GeoTIFF:** an image file that knows exactly where on Earth it belongs (coordinates embedded) — opens correctly in GIS software.
- **Evidence packet / provenance:** the machine-readable record of *which tool or model produced each claim*, including file fingerprints (SHA-256 hashes) so the exact software version is provable.

- **Withholding:** when evidence is too weak, the system says "insufficient evidence" instead of guessing — a designed behavior, not a failure.
- **Two model seats:** one adapted model trained by us on satellite benchmarks (the expert), one frozen general model (the neutral voice). Keeping them separate — and labeled — means the narrator can't silently drift from the evidence.